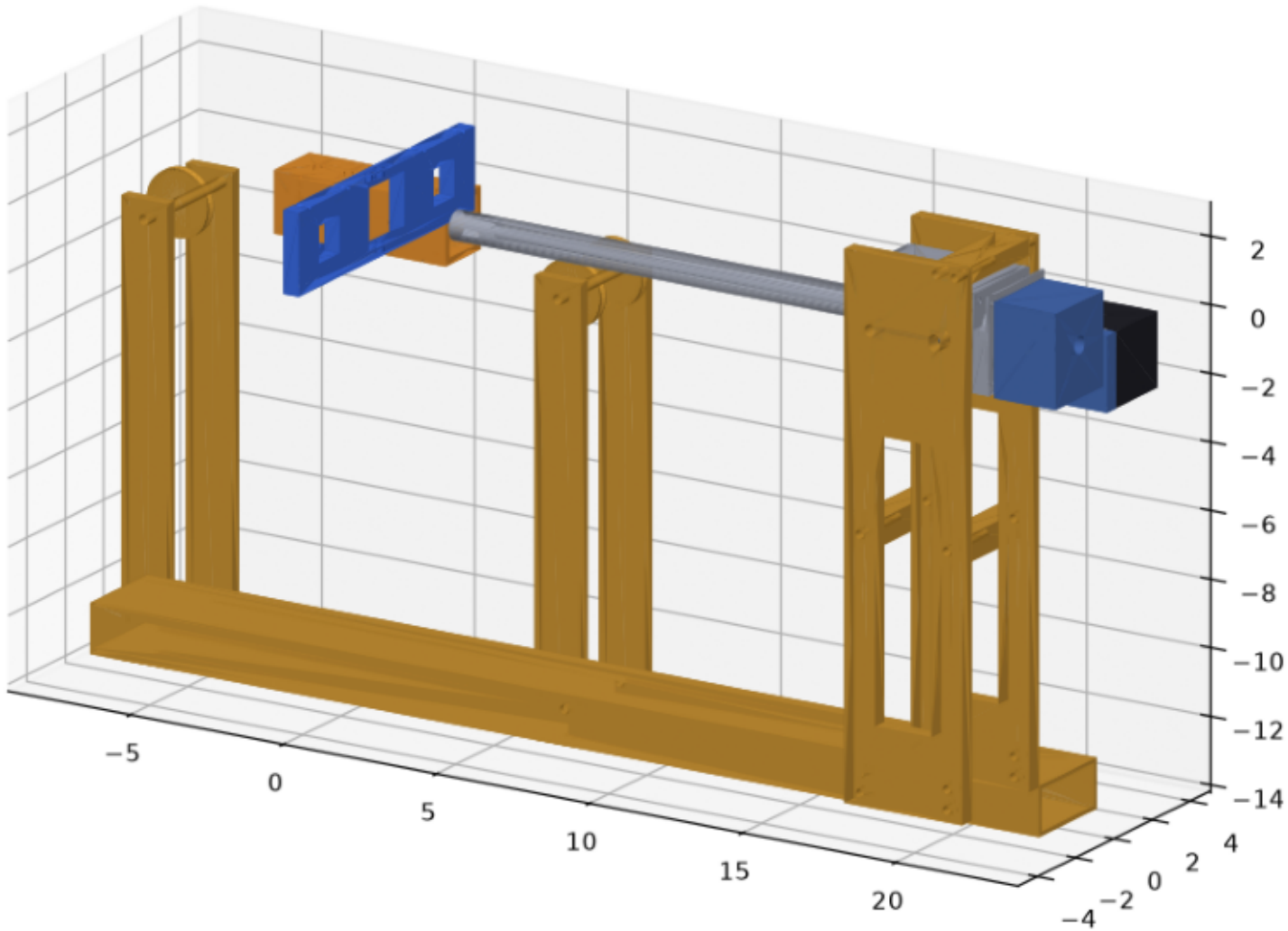


CALIBRATION ON THE ROLL SPINDLE (CAD RENDER, NTS)

6-component calibration on the roll spindle: square sleeve, roll arm, yoke at x = -2, drag/thrust pulleys



NOTES:

1. THE BALANCE IS CALIBRATED ON THE REAL ROLL SPINDLE: THE HOUSING PIVOTS ON ITS FRONT STUD BETWEEN THE HEAD PLATES; THE REAR STUD SITS IN THE LEVELING SLOT; RE-LEVEL WITH THE JACK SCREW AFTER EVERY LOAD.
2. SLEEVE REPLACES THE PUCK. YOKE AT ROLL 0/90/180/270, STIRRUP AT 45/135/315, ROLL ARM AT 0/180.
3. BENCH MODE ROLL RANGE -90..+180 DEG. SCHEDULE AND FIT: analysis/balance_calibration_fit_6c.py.

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES

TOLERANCES:

.X ±.03 .XX ±.010

.XXX ±.005 .XXXX ±.0005

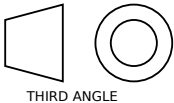
ANGLES ±0°30'

BREAK SHARP EDGES .005-.015

SURFACE FINISH 63 µin Ra MAX

INTERPRET PER ASME Y14.5-2018

UNDIMENSIONED GEOMETRY PER STEP MODEL



PARTS LIST

ITEM	PART / DWG NO.	DESCRIPTION	QTY	MATERIAL / SOURCE
1	WT-6C-301	Sleeve, calibration	1	6061-T6
2	WT-6C-302	Arm, roll calibration	1	6061-T6
3	WT-6C-303	Stirrup, 45 deg	1	6061-T6
4	WT-6C-304	Yoke, calibration	1	6061-T6
5	WT-6C-305	Plate, head	2	6061-T6
6	WT-6C-306-1	Bridge, jack	1	6061-T6
7	WT-6C-306-2	Spacer, cross	2	6061-T6
8	WT-6C-307	Tube, base	1	6061-T6 tube
9	WT-6C-308	Upright, pulley	4	6061-T6
10	WT-6C-309	Axle, pulley	2	416 SS ground
11	WT-6C-310	Pulley, cable, 2 in	3	6061-T6
12	-	Bearing R5-2Z (5/16 x 1/2)	3	bought
13	-	Shaft collar 5/16	4	bought
14	WT-6C-200	Roll spindle assy (with balance + sting)	1	see WT-6C-200
15	-	Stud 3/8 plain + nut (housing pivot/level)	4	bought
16	-	Jack screw 1/4-28 brass tip	1	bought
17	-	Ground pin 1/4 (load pin), dowels 1/4	3	bought
18	-	Cable 1/16 SS, eyebolt 1/4-20, hooks	1	bought
19	-	Dead weights, calibrated (Class F)	1	bought
20	-	SHCS 1/4-20 + nuts	16	bought

PROJECT				DRAWN / DATE			
Wind tunnel - motorized 6-component sting balance				Claude (AI-generated) 2026-10-02			
TITLE				CHECKED		APPROVED	
ASSEMBLY, 6-COMPONENT CALIBRATION							
MATERIAL				FINISH			
SEE PARTS LIST				SEE PART DRAWINGS			
DWG NO.		REV	SIZE	SCALE		WEIGHT	SHEET
WT-6C-300		A	B	NTS		-	3 OF 35